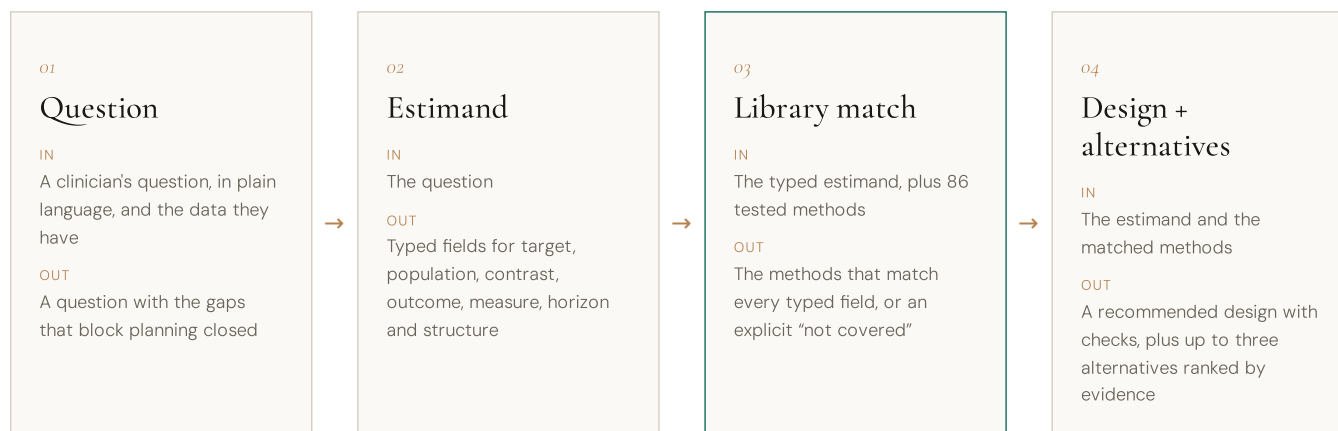


From a question to a *design*

Mallard does not start by choosing a study design. It first pins down exactly what the question is asking for, as a typed estimand. It then looks for methods in a tested code library that can estimate that quantity. Only after that does it recommend a design and set out the alternatives.

THE FOUR STAGES



The sections below follow one question through each stage. It is Mallard's own worked example, and the library results are read from the code library's metadata.

01 Start with the question

A clinician states what they want to know and what data they have. They do not need to name a design or a method.

Does scheduling a follow-up appointment before discharge change 30-day unplanned readmission?

Data: discharge and scheduling records from a hospital medicine service.

WHAT HAPPENS HERE

- A readiness check lists what is still missing before a plan can be written. Anything the clinician has already answered is dropped from the list.
- Mallard asks sharpening questions only when the answer would change the plan. For example: *does the scheduling system record when an appointment was created, or only when it takes place?*

02 Define the estimand

The question is turned into an estimand: the exact quantity the study will estimate. Every field has a fixed vocabulary, so the rest of the pipeline can check it with code instead of interpreting prose.

target associational	population Adults discharged alive from hospital medicine	exposure Follow-up scheduled before discharge	comparison No appointment scheduled before discharge
outcome Time to first unplanned readmission	outcomeType timeToEvent	summaryMeasure hazardRatio	timeHorizon 30 days after discharge
framing estimation	assignment nonrandomized	clustered false → independent sampling	repeatedMeasures false
competingEvents true (death within 30 days)			

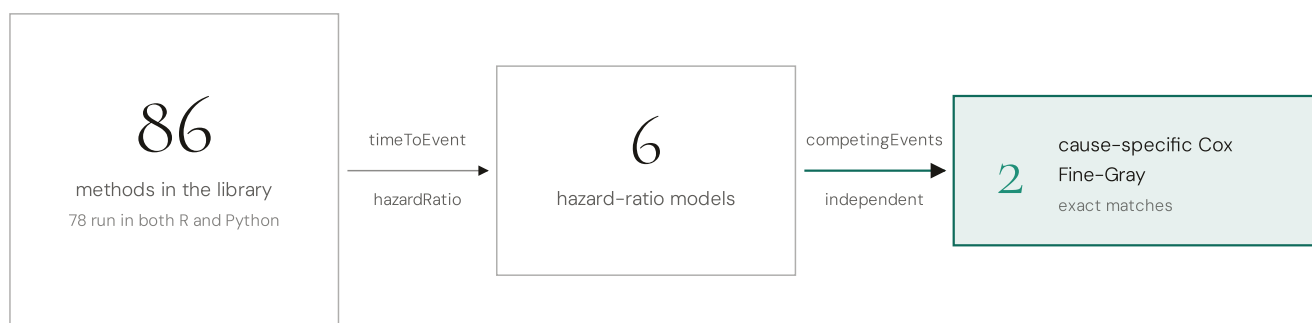
☐ Shaded fields feed the library gate in stage O3. Each must match the method's metadata exactly.

WHY THIS COMES FIRST

- Competing events is a single yes/no field, and it changes which methods are valid. A standard Cox model treats a patient who died as simply censored, which overstates readmission.
- Later steps check themselves against this estimand. If the analysis plan drifts on the target, the summary measure, the framing, the assignment or the clustering, the drift is flagged and the sample size is withheld.

03 Match the code library

The Mallard code library has 86 analysis methods. Each has R and/or Python code that has been run against a fixture dataset and checked against expected output. Each method also declares the estimands it is valid for. A method qualifies only if it accepts every gate field of the estimand.



Each arrow applies one set of gate fields. The first two fields narrow 86 methods to 6. The competing-events and sampling fields remove the standard Cox model and the case-cohort, nested case-control and time-varying variants, leaving 2.

THE GATE, FIELD BY FIELD

Library entry	target associational	outcomeType timeToEvent	summaryMeasure hazardRatio	competingEvents true	sampling independent	Result
Cause-specific Cox <code>cause-specific-cox</code>	✓	✓	✓	✓	✓	MATCH · R + PY
Fine-Gray subdistribution <code>fine-gray-subdistribution</code>	✓	✓	✓	✓	✓	MATCH · R
Cox proportional hazards <code>cox-proportional-hazards</code>	✓	✓	✓	✗ false only	✓	EXCLUDED
Case-cohort weighted Cox <code>case-cohort-weighted-cox</code>	✓	✓	✓	✗ false only	✗ caseCohort	EXCLUDED
Kaplan–Meier + log-rank <code>kaplan-meier-logrank</code>	✓	✓	✗ other	✗ false only	✓	EXCLUDED

Exact, not nearest

Methods are not scored or ranked. An entry either accepts every gate field or it is excluded. The Cox model matches on four of five fields and is still excluded.

No substitution

If no entry qualifies, the plan says the method is *not covered*. It does not quietly swap in a different estimand that the library happens to have.

Code with provenance

The code shown is taken from the matched entry at a pinned library commit. It is labelled as adapted and not yet run on the user's data.

04 Recommend a design, show the alternatives

The recommended design is the one that answers the estimand with the data the clinician already has, using a method the library covers. The alternatives are shown next to it, each with what it costs and what it would let you claim.

RECOMMENDED

ADJUSTED ASSOCIATION

MONTHS

EXISTING RECORDS

Retrospective cohort from discharge and scheduling records, with death as a competing risk

The exposure is already recorded at discharge, and the outcome is recorded in the same system. A cohort can therefore answer the question from data that already exists, without paying to collect new data. Because the two groups differ systematically, the analysis adjusts for confounders, and the result is reported as an association rather than an effect.

PRIMARY ANALYSIS · FROM THE LIBRARY

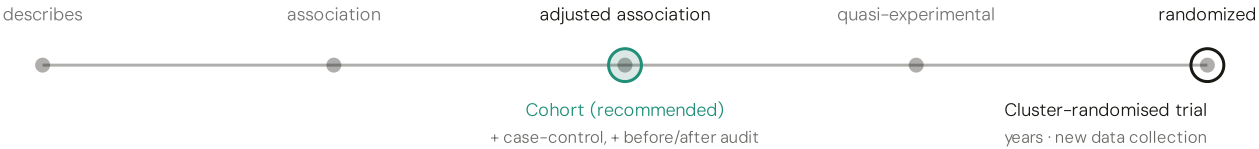
Cause-specific Cox regression, reported as an adjusted hazard ratio

COMPANION · FROM THE LIBRARY

Fine-Gray regression for the cumulative incidence of readmission

DESIGN-FIT CHECKS TO RUN BEFORE COMMITTING

- 1 At least 95% of appointments have a creation time before the discharge time.
If not: restrict to reliable timestamps, or switch to the case-control design.
- 2 Deaths recorded in the system are within 10% of an external death count.
If not: report the cause-specific hazard only.
- 3 At least 10 readmissions per covariate in the adjustment set.
If not: widen the record window or reduce to a minimal sufficient set.



Three of the four designs support the same claim, an adjusted association. They differ in where the exposure data comes from and what can go wrong. Only allocating the change at random turns the result into a causal effect.

ADJUSTED ASSOCIATION

ABOUT A YEAR

Case-control, exposure abstracted from charts

WINS WHEN

The scheduling timestamp can't be trusted, and chart review is affordable for a sample of about 600 records but not for everyone.

ADJUSTED ASSOCIATION

MONTHS

Before-and-after audit around the scheduling change

WINS WHEN

The go-live date is known and records on both sides already exist. It describes a change around that date and makes no claim that the scheduling change caused it.

RANDOMIZED

YEARS

Cluster-randomised trial by ward

WINS WHEN

The scheduling change can be assigned to wards instead of observed. This is the only design here that can answer the question as a causal effect.

WHY THIS ORDER

Choosing a design before the estimand is fixed invites a familiar method that answers a slightly different question. Here, a standard Cox model would ignore deaths. Fixing the estimand first, and accepting only exact library matches, means the recommended design comes with analysis code that has been run and checked for the question actually asked.

Worked example adapted from [Ask Mallard](#) and its open code library.